Adil Sukumar

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EDUCATION

Vellore Institute of Technology **Aug'25–May'30**
Integrated M.Tech. Artificial Intelligence & Bioinformatics | CGPA: 9.10/10

EXPERIENCE

British Airways **Oct'26–Present**
Data Science Job Simulation
- Undertake a self-paced virtual data science simulation focused on **customer insights and predictive analytics** for airline business decisions.

VIT Bhopal University **Jul'26–Present**
Student Researcher (Part-time)
- Perform sample preparation, **Gram staining**, **microscopy** and **foundational tissue-culture** procedures under faculty mentorship.
- Maintain **aseptic conditions**, manage laboratory equipment and document experimental results.

Pawsitive Diagnostics **Feb'26–Present**
Founder and CEO
- Lead **software, hardware and machine learning** development for a **wearable animal-health system**.
- Integrate motion, acoustic, ambient and surface-temperature sensors into edge-to-cloud workflows.
- Develop **signal filtering**, **individual baselines**, **activity recognition** and **context-aware anomaly alerts**.

ElectraWireless **May'26–Aug'26**
Lead Intern, IoT & Embedded Systems
- Co-created **ELLY**'s offline smart-home platform; implemented local **Web-LLM** voice control and **TensorFlow.js/MobileNet** appliance-scanning and facial-recognition workflows.

Sequensolutions **Dec'25–Mar'26**
AI and Bioinformatics Intern
- Used **AWS S3 and EC2** for sequencing-data storage and cloud-based processing.
- Worked on **variant-calling and RNA-seq** workflows using **Python and Linux** tools.
- Analyzed SCLC sequencing data through **read quality control**, **GRCh38 alignment** and **BAM processing**.
- Prioritized variants with **functional annotation** and **consequence filters**; documented interpretation limits.

PROJECTS

SCLC Somatic Variant Analysis | BWA, SAMtools, GATK, SnpEff **Dec'25–Mar'26**
- Analyzed public sequencing data from accession ERR6473446 through **read quality control**, **GRCh38 alignment**, **BAM processing** and **variant analysis**.
- Used **functional annotation** and **consequence filters** to prioritize **candidate variants**; documented pipeline choices and interpretation limits.

Zebrafish Formaldehyde Toxicity | Python, OECD QSAR Toolbox **2026**
- Analyzed 96-hour mortality in **30 adult zebrafish** using Python, **Abbott correction** and **log-concentration/probit interpolation** with **endpoint sensitivity checks**.
- Integrated **six OECD QSAR Toolbox/literature reports**, source hashes and chemical depictions with tested JSON/SVG analysis outputs.

TemperatureSense AI | ESP32-C3, Python, SHT40, NTC **Jul'26–Present**
- Developed baseline, filtering and **context-aware alert logic** for **surface-temperature and environmental sensor streams**.

TECHNICAL SKILLS

Bioinformatics: SRA Toolkit, FastQC, MultiQC, BWA, SAMtools, GATK, SnpEff/SnpSift
Programming and Data: Python, C++, SQL, NumPy, Pandas, SciPy, Linux
Computational Research: OECD QSAR Toolbox, variant analysis, RNA-seq workflows, sensitivity analysis
ML and Biosensing: scikit-learn, LightGBM, ONNX, ESP32, filtering, anomaly detection
Wet Lab: Gram staining, microscopy, aseptic technique, sample preparation and foundational tissue culture.
Cloud and Reproducibility: AWS S3/EC2, Git, GitHub Actions, Docker

LEADERSHIP AND ACHIEVEMENTS

WiCyS Student Chapter, VIT — Secretary | Dec'25–Present
- CyberShield Hackathon 2025, VIT — **First Place**.
- SKITECH National Innothon 3.0, Indore — **First Runner-Up**, national level.
- SASI National Hackathon 2026, Andhra Pradesh — **Top 6**, national level.